

SWH
50/75/80/99/100kW
Three Phase Hybrid Inverter (HV)

MODEL

SWH050KH-T1-L-G2 | SWH075KH-T1-G2

SWH080KH-T1-G2 | SWH099KH-T1-G2

SWH100KH-T1-G2

Optimized Performance
- 8 MPPTs,200% PV oversizing
- Up to 21A per string
- 110% Unbalanced Load
- 150% AC Overloading @10s
- DC Coupling for Smarter Energy and Higher Efficiency

Flexible & Adaptable Applications
- Dual Independent Battery Inputs for More Flexible Configuration, Supporting up to 220A
- Generator Start-Stop Control via DI/DO Interfaces
- Seamless Backup Power with up to 200A Output

Superb Safety & Reliability
- <10ms UPS-Level Switching with STS
- Type I + II SPD on DC side[1]
- Smart temp. sensing on AC and PV connectors
- Smart DC Circuit Breaker[1]

1: Optional functions or devices are purchased separately.

Model	SWH050KH-T1-L-G2	SWH075KH-T1-G2	SWH080KH-T1-G2	SWH099KH-T1-G2	SWH100KH-T1-G2
Battery Side					
Battery Type*1	Li-Ion				
Nominal Battery Voltage (V)	400	600			
Voltage Range (V)	300 ~ 600	300 ~ 800			
Start-up Voltage (V)	300				
Number of Battery Inputs	2				
Max. Continuous Charging Current (A)	85 × 2	85 × 2	100 × 2	110 × 2	110 × 2
Max. Continuous Discharging Current (A)	85 × 2	85 × 2	100 × 2	110 × 2	110 × 2
Max. Charging Power (kW)	50	75	88	99.99	110
Max. Discharging Power (kW)	50	75	88	99.99	110
PV Side					
Max. Input Power (kW)	100	150	160	200	200
Max. Input Voltage (V)*2	850	1000	1000	1000	1000
MPPT Operating Voltage Range (V)*3	160 ~ 700	160 ~ 950	160 ~ 950	160 ~ 950	160 ~ 950
Start-up Voltage (V)	200				
Nominal Input Voltage (V)	420	620	620	620	620
Max. MPPT Current (A)	42 × 8				
Max. MPPT Short Circuit Current (A)	55 × 8				
Number of MPPTs	8				
Number of Strings per MPPT	2				
AC Side (On-grid)					
Nominal Power (kW)	50	75	80	99.99	100
Max. Power (kW)	50	75	88*5	99.99	110*5
Nominal Apparent Power from/to Grid (kVA)	50	75	80	99.99	100
Max. Apparent Power to Grid (kVA)	50	75	88*6	99.99	110*6
Max. Apparent Power from Grid (kVA)	50	75	88*6	99.99	110*6
Nominal Voltage (V)	127/220,3L/N/PE	220 / 380, 230 / 400, 240 / 415, 3L / N / PE			
Voltage Range (V) (according to local standard)	114 ~ 139	180 ~ 280			
Nominal Frequency (Hz)	50 / 60				
Frequency Range (Hz)	45 ~ 55 / 55 ~ 65				
Nominal Current from / to Grid (A)	131.3	114.0 @380Vac; 108.3 @400Vac; 104.4 @415Vac	121.6 @380Vac; 115.5 @400Vac; 111.3 @415Vac	152.0 @380Vac; 144.4 @400Vac; 139.2 @415Vac	152.0 @380Vac; 144.4 @400Vac; 139.2 @415Vac
Max. Current to Grid (A)*7	131.3	114.0 @380Vac; 108.3 @400Vac; 104.4 @415Vac	133.8 @380Vac; 127.1 @400Vac; 122.5 @415Vac	152.0 @380Vac; 144.4 @400Vac; 139.2 @415Vac	167.2 @380Vac; 158.8 @400Vac; 153.1 @415Vac
Max. Current From Grid (A)*7	131.3	114.0 @380Vac; 108.3 @400Vac; 104.4 @415Vac	133.8 @380Vac; 127.1 @400Vac; 122.5 @415Vac	152.0 @380Vac; 144.4 @400Vac; 139.2 @415Vac	167.2 @380Vac; 158.8 @400Vac; 153.1 @415Vac
Power Factor	0.8 leading ~ 0.8 lagging				
THDI	<3%				
Back-up Side*4					
Nominal Output Apparent Power (kVA)	50	75	80	99.99	100
Max. Output Apparent Power (kVA)	50	75	88	99.99	110
Peak Output Power without Grid (kW)	120% @ 60s; 150% @ 10s	120% @ 60s; 150% @ 10s	9.1@220V 8.7@230V	12.1@220V 11.6@230V	110% @ continuous; 120% @ 60s; 150% @ 10s
Nominal Output Voltage (V)	127/220, 3L/N/PE	220 / 380, 230 / 400, 240 / 415, 3L / N / PE			
Nominal Output Frequency (Hz)	50 / 60				
Max. Output Current (A)	131.3	114.0 @380Vac; 108.3 @400Vac; 104.4 @415Vac	133.8 @380Vac; 127.1 @400Vac; 122.5 @415Vac	152.0 @380Vac; 144.4 @400Vac; 139.2 @415Vac	167.2 @380Vac; 158.8 @400Vac; 153.1 @415Vac
THDv (@Linear Load)	<3%				
On / Off-grid Switching Time	<10ms				
Generator Side*4					
Nominal Apparent Power (kVA)	50	75	80	99.99	100
Max. Apparent Power (kVA)	50	75	88	99.99	110
Nominal Voltage (V)	127/220, 3L/N/PE	220 / 380, 230 / 400, 240 / 415, 3L / N / PE			
Nominal Frequency (Hz)	50 / 60				
Frequency Range (Hz)	45 ~ 55 / 55 ~ 65				
Max. Current (A)	131.3	114.0 @380Vac; 108.3 @400Vac; 104.4 @415Vac	133.8 @380Vac; 127.1 @400Vac; 122.5 @415Vac	152.0 @380Vac; 144.4 @400Vac; 139.2 @415Vac	167.2 @380Vac; 158.8 @400Vac; 153.1 @415Vac
Efficiency					
Max. Efficiency	97.4%	98.1%	98.1%	98.1%	98.1%
European Efficiency	96.8%	97.7%	97.7%	97.7%	97.7%
Max. Battery to AC Efficiency	97.6%	98.2%	98.2%	98.2%	98.2%
MPPT Efficiency	99.9%				
Protection					
PV String Current Monitoring	Integrated				
PV Insulation Resistance Detection	Integrated				
Residual Current Monitoring	Integrated				
PV Reverse Polarity Protection	Integrated				
Battery Reverse Polarity Protection	Integrated				
Anti-islanding Protection	Integrated				
AC Overcurrent Protection	Integrated				
AC Short Circuit Protection	Integrated				
AC Overvoltage Protection	Integrated				
DC Switch	Integrated				
DC Surge Protection	Type II (Type I + II Optional)				
AC Surge Protection	Type II				
AFCI*8	Optional				
Rapid Shutdown	Optional				
Remote Shutdown	Optional				
General Data					
Operating Temperature Range (°C)	-35 ~ +60				
Relative Humidity	0 ~ 100%				
Max. Operating Altitude (m)	4000				
Cooling Method	Smart Fan Cooling				
User Interface	LED, LCD (Optional), WLAN + APP				
Communication with BMS	CAN				
Communication	RS485, WiFi + LAN + Bluetooth, 4G + Bluetooth (Optional)				
Communication Protocols	Modbus-RTU, Modbus-TCP				
Weight (kg)	97				
Dimension (W × H × D mm)	995 × 758 × 358				
Noise Emission (dB)	<60				
Power Self-consumption at Night (kW)	<15				
Ingress Protection Rating	IP66				
Anti-corrosion Class	C4				
Mounting Method	Wall Mounted				

*1: The Li-Ion battery usually contain two mainstream type: LFP and Ternary Lithium battery.
*2: For SWH075KH-T1-G2/SWH080KH-T1-G2/SWH100KH-T1-G2, when the input voltage ranges from 950V to 1000V, the inverter will enter the standby mode, and the voltage returns to 950V to enter the normal operation state.
*3: Please refer to the user manual for the MPPT Voltage Range at Nominal Power.
*4: The STS Box or STS Cabinet is needed.
*5: For Chile, Max. Power (kW): SWH080KH-T1-G2: 80kW, SWH100KH-T1-G2: 100kW.
*6: For Chile, Max. Apparent Power to Grid (kVA)/Max. Apparent Power from Grid (kVA): SWH080KH-T1-G2: 80kVA, SWH100KH-T1-G2: 100kVA.
*7: For Chile, Max. Current to Grid (A)/Max. Current from Grid (A): SWH080KH-T1-G2: 121.6@380Vac, 115.5@400Vac, 111.3@415Vac; SWH100KH-T1-G2: 152.0@380Vac, 144.4@400Vac, 139.2@415Vac.
*8: For Brazil, AFCI: Integrated.
*: Please visit SKYWORTH website for the latest certificates.

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